

Data Flow Diagram

Date	29 june 2026
Team ID	xxxxxx
Project Name	Credit card approval prediction
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

1. External Entities (Oval / Rounded Shapes)

These represent the human users or independent actors outside the boundary of the automated system that send or receive data.

- **Applicant / Bank Credit Officer:** The primary user who interacts with the front-end layout interface. They fill out demographic inputs and trigger the evaluation pipeline to check eligibility.
- **System Administrator / Data Engineer:** The technical actor responsible for feeding new historical source records (`application_record.csv` and `credit_record.csv`) into the training layer to periodically re-optimize the model parameters.

2. Processes (Rectangles with Numbered Headers)

These represent the specific programmatic actions that intercept, transform, clean, or evaluate data streams.

- **Process 1: Data Integration & Cleaning (Preprocessing)**
 - *Description:* Reads raw profiling tables, removes exact duplicate row signatures, maps credit status lists to clean structural binary indices (0 or 1), and fills out missing entries in categorical columns (like `OCCUPATION_TYPE`).
- **Process 2: Model Training & Evaluation**
 - *Description:* Splits the balanced dataset into training and validation folds, trains the Random Forest ensemble architecture with balancing weights, evaluates prediction diagnostics, and saves the binary package (`model.pkl`).
- **Process 3: User Request Capture & Encoding**
 - *Description:* Intercepts form data submitted through the Flask web interface (`app.py`), converts inputs into numeric vectors matching the exact schema requirements, and readies them for prediction.
- **Process 4: Eligibility Prediction & Rendering**
 - *Description:* Unpickles the production model, feeds the processed input vectors through the classifier rules, derives a target prediction flag (0 or 1), and returns a clear text verdict to the response page (`result.html`).

3. Data Stores (Rectangles with Solid Fill, No Numbers)

These represent places where information is stored for later application use.

- **Source Data Files (application_record.csv / credit_record.csv):** The primary CSV data storage blocks containing historical credit history and demographic details.
- **Trained Model Binary (model.pkl):** The persistent database file that holds the fully optimized, saved Random Forest mathematical weights for instant real-time predictions.

4. Data Flows (Labeled Arrows Connecting the Symbols)

This traces exactly how your variables pass from one block to another across the complete project pipeline.

- Raw CSV Entries → Flowing from **Source Data Files** into **Process 1 (Preprocessing)**.
- Clean Consolidated Dataframe → Flowing from **Process 1** into **Process 2 (Model Training)**.
- Optimized Weights Data → Flowing from **Process 2** to be saved inside the **Trained Model Binary (model.pkl)**.
- Raw Client Demographics → Flowing from the **Applicant/Credit Officer** into **Process 3 (Request Capture)** via the HTML form submit action.
- Encoded Feature Vector → Flowing from **Process 3** into **Process 4 (Eligibility Prediction)**.
- Model Rules Query → Flowing bidirectionally between **Process 4** and the **Trained Model Binary (model.pkl)** to run data points through the rules.
- Final Binary Output & Status String → Flowing from **Process 4** back to the **Applicant/Credit Officer** to print the final "Approved" or "Rejected" verdict screen.

Level 1 Data Flow Diagram: Credit Card Approval Prediction System

